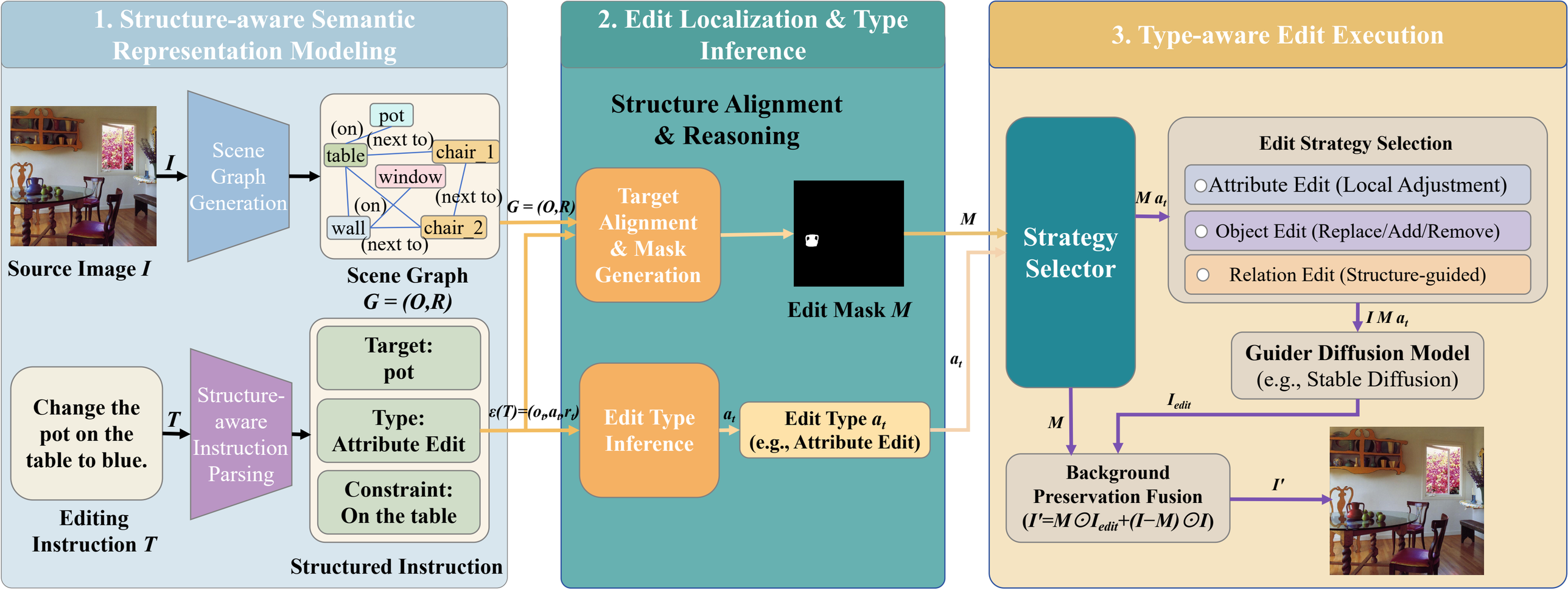
1. Structure-aware Semantic Representation Modeling
Source Image I
I
Scene Graph Generation
pot
(on)
(next to)
table
chair_1
window
(next to)
(on)
wall
chair_2
(next to)
Scene Graph
G = (O,R)
Change the pot on the table to blue.
Editing Instruction T
T
Structure-aware Instruction Parsing
Target: pot
Type: Attribute Edit
Constraint: On the table
Structured Instruction
ε(T)=(o_p,a_p,r_p)
2. Edit Localization & Type Inference
Structure Alignment & Reasoning
G = (O,R)
Target Alignment & Mask Generation
Edit Mask M
Edit Type Inference
a_t
Edit Type a_t (e.g., Attribute Edit)
M
a_t
3. Type-aware Edit Execution
Strategy Selector
M a_t
Edit Strategy Selection
Attribute Edit (Local Adjustment)
Object Edit (Replace/Add/Remove)
Relation Edit (Structure-guided)
I M a_t
Guider Diffusion Model
(e.g., Stable Diffusion)
I_edit
M
Background Preservation Fusion
(I'=M⊙I_edit+(I−M)⊙I)
I'